N-Tuple Collocation (NC)

Robust Reference-Free Error Covariance Estimation

Beyond Triple Collocation (TC) and Three-Cornered Hat (TCH)

Input

Collocated geoscientific datasets

No reliable reference truth

$x_1, x_2, \ldots, x_p$

TC

Exactly 3 datasets

Uncorrelated errors

Negative variance risk

TCH

Pairwise differences

Requires signal cancellation

Scaling-sensitive

Covariance-level problem

$$\mathbf{Q} = \sigma_y^2 \mathbf{a}\mathbf{a}^{\mathrm{T}} + \mathbf{\Sigma}_e$$

Estimate Σ_e without y

Proposed NC

Constrained covariance decomposition

$$\mathbf{Q} = \theta\theta^{\mathrm{T}} + \mathbf{\Sigma}_e$$

Fit a rank-one signal component

by minimizing off-diagonal residuals

$$\min_\theta \sum_{i \neq j} |Q_{ij} - \theta_i \theta_j|$$

$$\text{subject to} \quad \theta_i^2 \leq Q_{ii}$$

Then enforce residual PSD feasibility

Applications: three or more products

Output 1

Error covariance

without reference

Output 2

Feasible estimates

PSD residual covariance

Output 3

Data merging

Conditioned WA

Reference-free uncertainty characterization and multi-source data merging

Assumes signal-uncorrelated errors; physical error covariance is not uniquely identified